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Theory Assignment

Q. 1 (a) Define Artificial intelligence and list major application areas.

→ Definition of Artificial Intelligence.
Artificial intelligence is a branch of Computer science that aims to delop machine or Computer system that can perform tasks requiring human-like intelligence, such as learning, reasoning, problem-solving and decision-making

Application of AI:-
1. Healthcare:- disease diagnosis and medical assitance
2. Robotics:- intelligent robots for different tasks
3. Natural language processing - Understanding human language
4. Computer Vision - recognising and analysing images and videos.
5. Expert system:- providing solution using stored knowledge.

Q.1(b) Describe the history and evolution of Artificial Intelligence.

→

Year	Milestone
1950	Alan Turing propared the Turing
1956	The term Artificial Intelligence was introduced at the Dartmouth Conference.
1960s	Early AI program and problem-solving system were developed
1980s	Expert system becomes popular in industries.
1997	IBM Deep Blue defeated chess champion Garry kasparov.
2010s	Deep learning and neural network achieved major advan
2020s	Generative AI systems become widely used for text, image and other tasks.

Evolution of AI

1. Early AI : Rule-based system and logical reas
2. Expert system: Used expert knowledge to solve specific problems.
3. Machine Learning :- Machine learned patte from data.
4. Deep Learning :- Neural network handled Complex tasks wing large datasets.

Q-2(a) Define an intelligent agent. Explain agents, environment, and rationality with examples.

→ Intelligent Agent.

An intelligent agent is a system that percives its environment through sensors and perform action using actuators to achieve a goal.

Main Components :-

1. Agent : The system that makes desions.
2. Sensors : Used to Collect information from the environment.
3. Environment : The surrounding in which the agent operates.
4. Actuators : Used to perform actions.
5. Rationality : Choosing the best action based on available information to achieve the desired goal.

e.g. Self - Driving Car

- Agent : self - driving Car System
- Sensors : Cameras, GPS, and other Sensor
- Environment : Roads, vehicles, pedestrians and traffic signals.
- Actuators : steering brakes, and accelato
- Rational action : Applying brakes when a obstacle is detected.

Q.2 (b) Explain different types of AI environments and their influence an agent design.

→ An AI environment is the surrounding world in which an intelligent agent operates.

	Type	Meaning
1	fully Observable	Agent can access all relevent information.
2	Partially Observable	Some information is hidden or Unavaible.
3.	Deterministic	An action has a predictable
4.	Stochastic	Result involve uncertanity or randomness.
5.	Static	Enviroment does not change while the agent is deciding
6.	Dynamic	Environment can change during decision-making
7.	Discrete.	Limited, Countable state or actions.
8.	Continuous	State or action can vary Continuously.
9.	Single-Agent	Only one agent is involve
10.	Multi-Agent.	Multiple agen interac in the evvironment.

Q.3(a) Define a problem-solving agent. Explain steps involved in problem formulation with an example.

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A problem-solving agent is an agent that finds a sequence of action to reach a goal from an intial state.

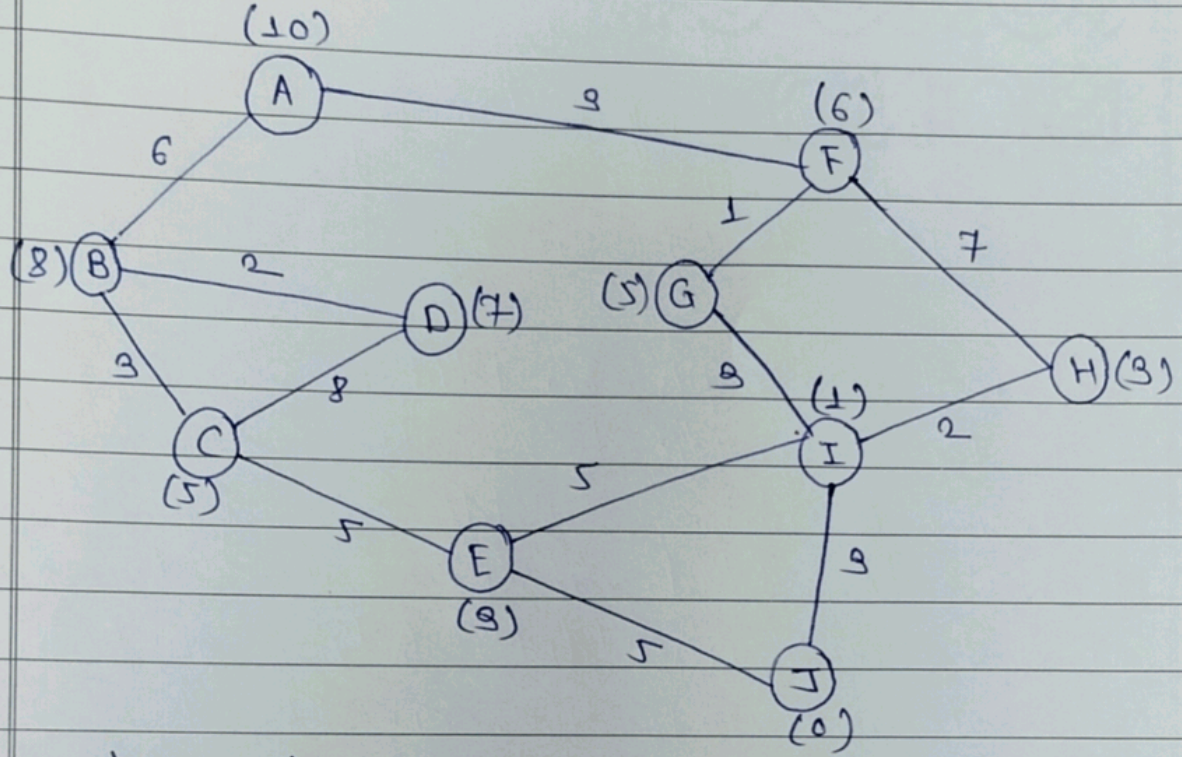
steps in problem formulation:-

1. Initial state: Starting point of the problem.
2. Goal state: Desired final result.
3. Action: Available action that the agent can perform.
4. Tranition Model:- Describes the result of applying an action.
5. Path Cost: Total cost of action taken to reach the goal.

Problem-Solving process.

1. Identify the intial state.
2. Define the goal state.
3. Identify passible action.
4. Explore the available paths.
5. choose a suitable path and reach the goal.

Q-3(b) find the most cost-effective path using the A* Algorithm.



∴ shortest path

∴ $f(n) = g(n) + h(n)$

$A \rightarrow B \quad 6 + 8 = 14$

$A \rightarrow F \quad 3 + 6 = 9$

$AF \rightarrow G \quad 3 + 1 + 5 = 9$

$\searrow H \quad 3 + 7 + 3 = 13$

$AFG \rightarrow I \quad 3 + 1 + 3 + 1 = 8$

$AFGI \rightarrow J \quad 3 + 1 + 3 + 3 + 0 = 10.$